

#### CH4 數列級數

1.(1)10 (2)576 2.42 3.3,12,21 4.1:2

5.5 6.5<sup>9</sup> 7.61529 8.見詳解 9.見詳解 10.9920

解析

$$1. (1) a_6 = a_1 + 5d = 50, a_{14} = a_1 + 13d = -62$$

可得  $d = -14, a_1 = 120$

$$a_n = a_1 + (n-1)d < 0 \Rightarrow 120 + (n-1) \times (-14) < 0$$

$$\Rightarrow 14n > 134 \Rightarrow n > \frac{134}{14} = 9. \dots$$

$n$  最小為 10  $\Rightarrow$  第 10 項開始為負

(2)  $S_n$  最大  $\Rightarrow$  前  $n$  項和最大  $\Rightarrow$  所求為  $S_9$

$$S_9 = \frac{(2a_1 + 8d) \times 9}{2} = \frac{[2 \times 120 + 8 \times (-14)] \times 9}{2} = 576$$

$$2. a_1 = 2, a_{10} = a_1 + 9d = 74 \Rightarrow d = 8$$

$$\therefore a_6 = a_1 + 5d = 2 + 5 \times 8 = \underline{42}$$

3. 令此三正數為  $a-d, a, a+d$

則  $(a-d) + a + (a+d) = 36$ , 得  $a = 12$

又  $12-d+1, 12+4, 12+d+43$  成等比數列

$$\therefore 16^2 = (13-d)(55+d) = 715 - 42d - d^2$$

$$\Rightarrow d^2 + 42d - 459 = 0 \Rightarrow (d+51)(d-9) = 0$$

、  $d = -51$  (不合) 或 9

故此三數為 3, 12, 21

4. 令此兩等差數列為  $\langle a_n \rangle$  與  $\langle b_n \rangle$ , 公差分別為  $d$  與  $e$

$$\text{則 } \frac{S_n}{S_n'} = \frac{\frac{n}{2}[2a_1 + (n-1)d]}{\frac{n}{2}[2b_1 + (n-1)e]} = \frac{2a_1 + (n-1)d}{2b_1 + (n-1)e} = \frac{3n+1}{7n-11}$$

$$\therefore \frac{a_7}{b_7} = \frac{a_1 + 6d}{b_1 + 6e} = \frac{2a_1 + 12d}{2b_1 + 12e} = \frac{3 \times 13 + 1}{7 \times 13 - 11} = \frac{40}{80} = \frac{1}{2}, \text{ 故 } a_7 : b_7 = 1 : 2$$

$$5. S_n = \frac{2\left(1 - \left(\frac{2}{3}\right)^n\right)}{1 - \frac{2}{3}} = \frac{422}{81} \Rightarrow 1 - \left(\frac{2}{3}\right)^n = \frac{211}{243}$$

$$\Leftrightarrow \left(\frac{2}{3}\right)^n = \frac{32}{243} = \left(\frac{2}{3}\right)^5 \therefore n = 5$$

$$6. \text{ 令 } S = \frac{2}{5} + \frac{4}{5^2} + \frac{6}{5^3} + \frac{8}{5^4} + \cdots + \frac{20}{5^{10}}$$

$$-\frac{1}{5}S = \frac{2}{\frac{5^2}{5} + \frac{4}{5^3} + \frac{6}{5^4} + \frac{8}{5^5} + \cdots + \frac{20}{5^{10}}} = \frac{2\left(1 - \left(\frac{1}{5}\right)^{10}\right)}{1 - \frac{1}{5}} - \frac{20}{5^{11}}$$

$$\begin{aligned} \frac{4}{5}S &= \frac{2}{5} + \frac{2}{5^2} + \frac{2}{5^3} + \frac{2}{5^4} + \cdots + \frac{2}{5^{10}} - \frac{20}{5^{11}} \\ &= \frac{1}{2}\left(1 - \frac{1}{5^{10}}\right) - \frac{20}{5^{11}} \therefore S = \frac{5}{8}\left(1 - \frac{1}{5^{10}}\right) - \frac{5}{5^{10}} = \frac{1}{8}\left(5 - \frac{9}{5^9}\right) \therefore k \\ &= \underline{5^9} \end{aligned}$$

$$7. 10000 \times 1.07 + 10000 \times 1.07^2 + 10000 \times 1.07^3 + \cdots + 10000 \times 1.07^5$$

$$= 10000 \times (1.07 + 1.07^2 + \cdots + 1.07^5)$$

$$= 10000 \times \frac{1.07 \times (1.07^5 - 1)}{1.07 - 1}$$

$$= 10000 \times \frac{(1.07)^6 - 1.07}{0.07} \approx 61529$$

$$8. (1) a_1 = 4, a_2 = \frac{11a_1 - 9}{4a_1 - 1} = \frac{35}{15} = \frac{7}{3},$$

$$a_3 = \frac{11a_2 - 9}{4a_2 - 1} = \frac{\frac{77}{3} - 9}{\frac{28}{3} - 1} = 2, a_4 = \frac{11a_3 - 9}{4a_3 - 1} = \frac{13}{7}$$

$$(2) \text{ 推测 } \underline{a_n = \frac{3n+1}{2n-1}}$$

$$(3) \textcircled{1} \text{ 當 } n = 1 \text{ 時, } a_1 = \frac{3 \times 1 + 1}{2 \times 1 - 1} = 4 \text{ 成立}$$

$$\textcircled{2} \text{ 設 } n = k \text{ 時, } a_k = \frac{3k+1}{2k-1} \text{ 成立}$$

則  $n = k + 1$  時,

$$a_{k+1} = \frac{11a_k - 9}{4a_k - 1} = \frac{11 \times \frac{3k+1}{2k-1} - 9}{4 \times \frac{3k+1}{2k-1} - 1} = \frac{\frac{33k+11-18k+9}{2k-1}}{\frac{12k+4-2k+1}{2k-1}} = \frac{15k+20}{10k+5} = \frac{3k+4}{2k+1} = \frac{3(k+1)+1}{2(k+1)-1} \text{ 成立}$$

故由數學歸納法得證，對任意正整數  $n$ ，

$$a_n = \frac{3n+1}{2n-1} \text{ 恆成立}$$

9. (1) 當  $n = 1$  時

$$10^1 + 3 \cdot 4^1 + 5 = 10 + 12 + 5 = 27 = 9 \times 3 \text{ 為 } 9 \text{ 之倍數 } \therefore \text{成立}$$

(2) 設  $n = k$  時本命題也成立，即  $10^k + 3 \cdot 4^k + 5 = 9t$ ， $t$  為正整數

當  $n = k + 1$  時

$$\begin{aligned} 10^{k+1} + 3 \cdot 4^{k+1} + 5 &= 10 \cdot 10^k + 3 \cdot 4 \cdot 4^k + 5 = 10(9t - 3 \cdot 4^k - 5) + 12 \cdot 4^k + 5 \\ &= 90t - 18 \cdot 4^k - 45 + 12 \cdot 4^k + 5 = 9(10t - 2 \cdot 4^k - 5) \text{ 為 } 9 \text{ 之倍數} \end{aligned}$$

$\therefore$  由數學歸納法得證，對所有自然數  $n$ ， $10^n + 3 \cdot 4^n + 5$  都可以被 9 整除

$$10. \text{ 一般項 } = k(k+1) = k^2 + k$$

$$\text{所求} = 1^2 + 1 + 2^2 + 2 + 3^2 + 3 + \cdots + 30^2 + 30 = (1^2 + 2^2 + \cdots + 30^2) +$$

$$(1 + 2 + 3 + \cdots + 30) = \frac{30 \times 31 \times 61}{6} + \frac{30 \times 31}{2} = 9920$$